SiMPL Electronic Batch Record — PI Sheet | **NEW (XStep-based EBR)**

AZD0543 — Virus Inactivation + Concentration & Diafiltration (2000 L, Process 1)

Material / Product	AZD0543 — Antibody-Drug Conjugate (DSP, Process 1, GPF, 2000 L)
Records covered	PN 8012441 Virus Inactivation (Low pH) + PN 8012475 Concentration & Diafiltration (Large Skid UF/DF)
MABR / MPR	MABR-0027570 (VI) | PN 8012475 (C&D)
Process Order No.	____________
Batch #	____________
Plant / Facility	GPF
Effective Date	____________

How to read this: the digital EBR is assembled from XStep building blocks. Each card shows the original MPR section it replaces ("Original BR: §___") and is tagged NEW (built new), VARIANT (a new variant of an existing SMPL XStep), or REUSE (reuses an existing DE1 100 SMPL XStep as-is — shown as a summary, no new mock-up). For the full line-by-line comparison (verbatim old text beneath each card) see the companion **"New vs Old"** PDF; traceability is in the RTM.

8 new + **0** variant + **29** reuse = **37** XSteps for both records.

XStep ↔ Original MPR Crosswalk

EBR Phase	XStep	Type	Original MPR Section
Process Order Information & Front Matter	SMPL: Order Header Details	R	Header
	SMPL: Signature Table	R	Personnel ID
	SMPL: Display BOM	R	§4
	SMPL: Room & Equipment Assign	R	§7 (VI 7.3-7.4 / C&D 7)
	SMPL: Additional Manufacturing Supplies	N	§4
	SMPL: Additional Solution Batches (Solution Switch)	N	§5
	SMPL: Process Notes & Global Limits (Acknowledgement)	R	§6
Reusable Building Blocks (instantiated across BOTH records)	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	N	VI 7.5 · C&D 7.4-7.6
	SMPL: Product Vessel Weigh	R	VI §7/8/15 · C&D §9/11/16/17/18
	SMPL: Calc — Three Columns (Value1 op Value2 = Result)	R	VI 7.12 / 8.10 / 14.15 · C&D §7/16/17
	SMPL: Multi-Variable Calculation (3-input / ranged)	R	C&D §12/16/17
	SMPL: Product Mixing	R	VI §15 · C&D §9/16/17
	SMPL: Product Sampling & DLIMS Submission	R	VI §15/Att1 · C&D §8/14/16/17/Att3
	SMPL: Product pH / Conductivity / Temperature	R	VI 7.1/15.5 · C&D 8.12/14.11/17
	SMPL: Hold Time Table	N	VI §19 (Att3) · C&D §9/§25 (Att5)
	SMPL: Product Recirculation Worksheet	R	VI Att6 · C&D Att7
	SMPL: Transfer Tubing / Hosing Information	R	VI Att5 · C&D Att1
	SMPL: Product Storage & Cross-Record	N	VI 15.8 · C&D §9/§18
	SMPL: Instruction & Sign-off	R	VI §8/16/20 · C&D §11/20
	SMPL: Yield Calculations	R	VI Att4 · C&D Att8
	SMPL: Comments / Deviations Section	R	VI Att7 · C&D Att9
	SMPL: Non-Routine Sampling Record	R	VI Att2 · C&D Att4
Virus Inactivation — PN 8012441 (Low pH, 2000 L)	SMPL: VI Treatment Vessel Setup	N	§8

	SMPL: Iterative pH Titration Table (Acid / Base)	N	§10 / 12 / 14
	SMPL: Incubation Timing & Incubated Sample (block stack)	R	§11 / 13
	SMPL: Starting / Transfer Temperature Check (block stack)	R	§7.14 / 9
	SMPL: Store VI Product & Hold-Time Link (block stack)	R	§15.8
Concentration & Diafiltration — PN 8012475 (Large Skid UF/DF, 2000 L)	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	R	§7.2 / §11 / Att6
	SMPL: Run Skid Recipe (block stack)	R	§8/10/13/15/19
	SMPL: Minimum Membrane Surface Area (block stack)	R	§7.7
	SMPL: Process Input Calculations (calc-block stack)	R	§12
	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	N	Att2 (§13-15)
	SMPL: Conditional Continued Diafiltration (block stack)	R	§14
	SMPL: Recovery Calculations & Execution (block stack)	R	§16
	SMPL: Dilution Decision & Calculations (block stack)	R	§17
	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	R	§18
	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	R	§19 / §20

Process Order Information & Front Matter

SMPL: Order Header Details REUSE

Original BR: Header

Product / PN, process order numbers, recorded by — standard SAP order header. Reused as-is.

Reuses: **DE1 100: Display Process Order Header Data (AZ Components) / SXS: SIMPL Order Header**

SMPL: Signature Table REUSE

Original BR: Personnel ID

Personnel identification: print name, signature, initials for everyone executing the record. Reused as-is.

Reuses: **DE1 100: SMPL Header: Signature Table**

SMPL: Display BOM REUSE

Original BR: §4

Bill of Materials — equipment (VI Treatment Vessel / UF-DF skid, meters, scale, mixer, FIT) and components / filters / bags with SAP consumption. Reused as-is.

Reuses: **DE1 100: Display BOM Material Table**

SMPL: Room & Equipment Assign REUSE

Original BR: §7 (VI 7.3-7.4 / C&D 7)

Record the process room and assign every instrument up front (pH/cond meter, thermometer, scale, mixer, stir plate, pump) with calibration check. Downstream steps retrieve via Get [Type] (SMPL: Equipment Select, which already carries Calibration Due Date). Reused as-is.

Reuses: **DE1 100: SMPL: Room/Equipment Assign + SMPL: Equipment Select (field-verified: has ID + Description + Calibration Due Date)**

SMPL: Additional Manufacturing Supplies NEW

Original BR: §4

Instructions

If additional materials (filter, bag, assembly, etc.) are used during processing, verify the material is acceptable per Section 4 and record its information; document the reason in the Comments Section. Enter the Part No. (Material Description auto-fills) and Batch No. (Exp. Date auto-fills). **SAP consumption posts automatically via the Goods Issue process message (Z_PICONS, movement type 261).** Built new: a variant of SMPL: Additional Assembly with an added MPR Step No.

Reuses: **Variant of DE1 100: SMPL: Additional Assembly (adds MPR Step No. / Material Description) — no known client variant**

Data Input

#	MPR Step No.	Part No.	Material Description	Batch No.	Exp. Date	Serial No.	Autoclave ID	Autoclave Exp.	Performed By *
1								⏱	

+ Add Row

Witnessed By *

SMPL: Additional Solution Batches (Solution Switch) NEW

Original BR: §5

Instructions

When a process solution must be replaced with a different batch during processing, record the switch: indicate the applicable step, confirm the new solution is the same Part No. as the one replaced, and record Part No. / Batch No. / Exp. Date. Document the reason in the Comments Section. Built new: a variant of SMPL: Material Consumption (which carries Part No. / Batch / Exp.) with an added Applicable Step No.

Reuses: **Variant of DE1 100: SMPL: Material Consumption (adds Applicable Step No.; field-verified NOT Solution Summary) — no known client variant**

Data Input

#	Applicable Step No.	Part Number	Batch Number	Exp. Date	Performed By *
1					

+ Add Row

Witnessed By *

SMPL: Process Notes & Global Limits (Acknowledgement) REUSE

Original BR: §6

Global process notes / limits (15-27 °C; 1 L = 1 kg; within-expiry; equipment-ID attests calibration; Targets/Alert Limits; NaOH corrosive). The operator acknowledges; each limit is enforced at the step that captures its parameter. Reused as-is (Text Instructions shell, content per MBR).

Reuses: **DE1 100: SXS: Text Instructions with Sign-off (notes authored in IV_INSTR; DE2 903: Long Text Instructions)**

Reusable Building Blocks (instantiated across BOTH records)

SMPL: Incoming Product Information (Weight / Concentration / DLIMS) NEW

Original BR: VI 7.5 · C&D 7.4-7.6

Instructions

Record the incoming / starting product for each vessel or bag: **tare + net weight** (kg = L), the **SoloVPE protein concentration** (g/L, sample per SOP-0107091) and the **DLIMS project / sample numbers** it is reported under. Built new — no existing block combines these. **Goods issue has no batch field → consume the upstream Affinity / Pod-Harvest PN/batch.**

Reuses: **NEW — no DE1 100 block combines weight + concentration + DLIMS (field-verified: Material Consumption has PN/Batch, Solution Summary has measurement, neither has both)**

Data Input

Vessel / Bag #	Tare Weight (kg=L)	Net Weight / Volume (kg=L)	SoloVPE Concentration (g/L)	DLIMS Project No.	DLIMS Sample No.	Performed By *
1						

+ Add Row

Witnessed By *

SMPL: Product Vessel Weigh REUSE

Original BR: VI §7/8/15 · C&D §9/11/16/17/18

Get Scale / Balance (ID + Calibration Due auto-populate), record gross weight; net computed (Gross − Tare, 1 kg = 1 L). Reused as-is.

Reuses: **DE1 100: SMPL: Record Scale Weight (Tare Weight / Select Vessel Type & Tare Weight variants)**

SMPL: Calc — Three Columns (Value1 op Value2 = Result) REUSE

Original BR: VI 7.12 / 8.10 / 14.15 · C&D §7/16/17

Two-operand calc archetype (Value 1 [× ÷ + −] Value 2 = Result) — e.g. VI 7.12 Net = Gross − Tare. Reused as-is.

Reuses: **DE1 100: SMPL: Calc Three Columns (2-input — field-verified: Input Val1 / Input Val2 / Result)**

SMPL: Multi-Variable Calculation (3-input / ranged) REUSE

Original BR: C&D §12/16/17

Multi-operand calc archetype — 3-input (Three Variable Calc) or ranged result (Calc Range Values). Reused as-is.

Reuses: **DE1 100: SMPL: Three Variable Calc (3-input) / SMPL: Calc Range Values (ranged min/target/max)**

SMPL: Product Mixing REUSE

Original BR: VI §15 · C&D §9/16/17

Get Mixer (ID + Description), record agitation RPM, stamp start/end; duration computed. Reused as-is.

Reuses: **DE1 100: SMPL: Mixing Time (field-verified: Agitation RPM + Start/End + Duration of Mixing)**

SMPL: Product Sampling & DLIMS Submission REUSE

Original BR: VI §15/Att1 · C&D §8/14/16/17/Att3

Remove the sample per Att 1/3 & SOP-0107097; aliquot & label per SOP-0107056; submit to DLIMS per SOP-0080295; stamp date/time; record DLIMS project / sample numbers. Reused as-is.

Reuses: **DE1 100: SMPL: Sampling Record + SMPL: Sample Submission Chart**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Sampling Record — pull & aliquot (SOP-0107097 / label SOP-0107056)

Sample Designation *

▸ Record

Sampling Date & Time *

Sample Volume *

Intermediate Storage Temp (°C) *

SMPL: Sample Submission Chart — submit to DLIMS (SOP-0080295)

#	DLIMS Test Name	DLIMS Project No.	DLIMS Sample No.	Tube / Container Type	Location	Submitted?	Performed By *
1						Yes	

+ Add Row

SMPL: Product pH / Conductivity / Temperature REUSE

Original BR: VI 7.1/15.5 · C&D 8.12/14.11/17

Get pH/Cond Meter + Get Thermometer, then record range-validated pH / conductivity / temperature per reading. Assembled from two Equipment Selects + the Solution Summary results block (collapses per row). Contact supervisor if out of specification.

Reuses: **DE1 100 stack: SMPL: Equipment Select ×2 (pH/Cond Meter + Thermometer) + SMPL: Solution Summary - Data Recording (range-validated; field-verified = Equipment + Min/Max Target + Actual Value + Cond High/Low)**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Equipment Select — pH / Conductivity Meter

▸ Get pH / Cond Meter

pH / Cond Meter ID

Description

Calibration Due Date

SMPL: Equipment Select — Thermometer

▸ Get Thermometer

Thermometer ID

Description

Calibration Due Date

SMPL: Solution Summary - Data Recording — range-validated readings (collapses per row)

#	Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High / Low)
1	pH					

+ Add Row

SMPL: Hold Time Table NEW

Original BR: VI §19 (Att3) · C&D §9/§25 (Att5)

Instructions

Track product hold time across storage moves. Per hold: select the storage condition, stamp start and end; RT and 2-8 °C durations computed separately. Totals computed — alert if RT total exceeds 72 h or the combined total exceeds 168 h (contact supervisor). Built new: no block has the duration + threshold logic.

Reuses: **NEW — no DE1 100 block carries the RT / 2-8 °C hold DURATIONS + >72 h / >168 h threshold logic (Solution Summary is measurement, Solution Final Storage is a location)**

Data Input

Hold #	Location	Storage Temp		Storage Start		Storage End	RT Hold (h)	2-8°C Hold (h)	Performed By *
1		2-8 °C	▸ Record		▸ Record				

+ Add Row

Total RT Hold (h)

Total 2-8°C Hold (h)

Total Combined Hold (h)

Witnessed By *

SMPL: Product Recirculation Worksheet REUSE

Original BR: VI Att6 · C&D Att7

Record each product recirculation: identify the applicable MPR step & product name, record the recirculation tubing information, then stamp mixing start and end; duration computed (target ≥ 40 min). Assembled from a Record Text + Material Consumption + the Timer block; was mis-flagged 'new' earlier.

Reuses: **DE1 100 stack: SMPL: Record Text Value (Applicable MPR Step / Product Name) + SMPL: Material Consumption (recirculation tubing Part/Batch/Exp/Autoclave) + SMPL: Timer for Start-End Process (field-verified: Start/End Date-Time + Time Difference)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Record Text Value — recirculation context

Applicable MPR Step *

Product Name *

SMPL: Material Consumption — recirculation tubing SAP Goods Issue · Z_PICONS mvt 261

Tubing Part No.	Batch No.	Exp. Date	Autoclave Cycle No.	Autoclave Exp.	Performed By *

SMPL: Timer for Start-End Process — mixing by recirculation (≥ 40 min)

▸ Record

Mixing Start Time *

▸ Record

Mixing End Time *

Mixing Duration (≥ 40 min)

SMPL: Transfer Tubing / Hosing Information REUSE

Original BR: VI Att5 · C&D Att1

Record each transfer tubing / flex hose: Part No., Batch No. (Exp. auto), autoclave cycle / expiry, hose serial, cleaning batch / expiry. Reused as-is.

Reuses: **DE1 100: SMPL: Material Consumption / SMPL: Additional Assembly (Part/Batch/Exp/Serial/Autoclave)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Material Consumption — transfer tubing (one row per line) SAP Goods Issue · Z_PICONS mvt 261

#	Line / Use	Tubing Part No.	Batch No.	Exp. Date	Autoclave Cycle No.	Autoclave Exp.	Performed By *
1	Feed Line					⏱	

+ Add Row

SMPL: Additional Assembly — flex hose (one row per line) SAP Goods Issue · Z_PICONS mvt 261

#	Line / Use	Hose Part No.	Hose Serial No.	Hose Cleaning Cycle No.	Hose Cleaning Exp.	Performed By *
1	Feed Line				⏱	

+ Add Row

SMPL: Product Storage & Cross-Record NEW

Original BR: VI 15.8 · C&D §9/§18

Instructions

Record the initial storage condition (RT / 2-8 °C) and stamp the storage start date/time, and cross-record the storage linkage into the Hold-Time attachment / next MPR. Built new: SMPL: Solution Final Storage carries a Storage Location but not the storage-condition + start-date/time this record needs.

Reuses: **Variant of DE1 100: SMPL: Solution Final Storage (adds storage condition RT/2-8 °C + start date/time + cross-record; block has Storage Location only) — no known client variant**

Storage Condition (RT / 2-8 °C) * 2-8 °C

▸ Record

Date *

Time *

Cross-Record — Step / Value written to next MPR *

Performed By *

Witnessed By *

SMPL: Instruction & Sign-off REUSE

Original BR: VI §8/16/20 · C&D §11/20

Reusable instruction-only shell (line/valve connections, dip-tube confirmation, N/A gates, product transfer, BPR review). Text authored per use in IV_INSTR; operator reads and signs. Reused as-is.

Reuses: **DE1 100: SXS: Text Instructions with Sign-off (DE2 903: Long Text Instructions)**

SMPL: Yield Calculations REUSE

Original BR: VI Att4 · C&D Att8

% Yield = Final Product Mass ÷ Load Mass × 100 (masses carry from the weigh / concentration steps). Reused as-is.

Reuses: **DE1 100: SMPL: Yield Calculations**

SMPL: Comments / Deviations Section REUSE

Original BR: VI Att7 · C&D Att9

Free-text log for comments and documented deviations (step no. + initials/date). Reused as-is.

Reuses: **DE1 100: SXS: Phase Comments (no clean 'SMPL: Comments' — closest confirmed component)**

SMPL: Non-Routine Sampling Record REUSE

Original BR: VI Att2 · C&D Att4

Ad-hoc samples requested by MEMO: process section, contents, container, quantity, designation, DLIMS. Reused as-is.

Reuses: **DE1 100: SMPL: Non-Routine Sampling Record**

Virus Inactivation — PN 8012441 (Low pH, 2000 L)

SMPL: VI Treatment Vessel Setup NEW

Original BR: §8

Instructions

Set up the VI Treatment Vessel. The **selection dropdowns below drive conditional activation** — each gate activates or deactivates the section beneath it, so they are all built into this one XStep. First decide whether the incoming Affinity product vessel is reused as the VI Treatment Vessel (if so, all of Section 8 is N/A). Otherwise obtain & label the vessel (AZD0543 / "VI Treatment Vessel" / Batch / Part / Initials / Date), complete the Bag *or* Tank section, record the vent + product filters, and capture the tare weight with the filters-attached confirmation. **Filters post via Goods Issue (Z_PICONS 261).**

Reuses: **NEW — bespoke single XStep. The Affinity-vessel / Bag / Tank / vent-filter / filters-attached dropdowns DRIVE conditional activation of the sections they gate, and in PI-PCS the activating dropdown must live in the SAME XStep as the sections it activates/deactivates (a dropdown XStep cannot reach down and gate a separate XStep below it). Fields field-verified against SMPL: Select Vessel Type & Tare Weight + SMPL: Material Consumption, but built as one XStep for the branching.**

Use Affinity Product Vessel as VI Treatment Vessel? * No — *activates / deactivates the section below*

Yes → all of Section 8 is N/A and every section below deactivates (continue at Section 9/10). No → obtain & label the vessel and complete the applicable sections below.

Bag Required? * No — *activates / deactivates the section below*

Bag Information (activated when Bag Required = Yes) SAP Goods Issue · Z_PICONS mvt 261

Bag Part No.	Batch No.	Exp. Date	Serial No.	Performed By *

Mixer Drive attached? * Yes / No

Top Base attached? * Yes / No

Magnetic Clamp attached? * Yes / No

Tank Required * No — *activates / deactivates the section below*

Tank Information (activated when Tank Required = Yes)

Tank ID	CIP Batch ID	CIP Exp. Date/Time	Pressure Test ID		Pressure Test Date	Pressure Test Time	SIP Batch ID
				▸ Get Pressure Date/Time			

Vent Filter Required * No — *activates / deactivates the section below*

Vent Filter Information (see Section 4; bold PNs require autoclaving) SAP Goods Issue · Z_PICONS mvt 261

Filter Position	Part No.	Batch No.	Exp. Date	Serial No.	Autoclave Cycle No.	Autoclave Exp.	Performed By *
Vent Filter (if applicable)							

Product Filter Information (see Section 4; bold PNs require autoclaving) SAP Goods Issue · Z_PICONS mvt 261

#	Filter Position	Part No.	Batch No.	Exp. Date	Serial No.	Autoclave Cycle No.	Autoclave Exp.	Performed By *
1	Product Filter 1							

+ Add Row

▸ Get Scale / Balance

Scale / Balance ID

Scale / Balance Calibration Due Date

Tare Weight of VI Treatment Vessel (kg = L) *

Product Filter attached when tare obtained? * Yes / No

Vent Filter attached when tare obtained? * Yes / No

No filters — vessel is a bag? * Yes / No

Recorded By / Date*

SMPL: Iterative pH Titration Table (Acid / Base) NEW

Original BR: §10 / 12 / 14

Instructions

Iterative pH adjustment by incremental buffer addition. Record initial pH / conductivity / temperature and start time. Per addition: gross weight before & after (net + running total computed), stamp the pH-sample time and record adjusted pH. Target — acid pH 3.5 (NOR 3.4-3.6), neutralization pH 7.2-7.6. Re-standardize the pH meter (4.0/10.0) once pH ≥ 4.0. Buffer consumption posts via Goods Issue (Z_PICONS 261) on the total net weight. Built new — bespoke titration table; the Get Scale / Get pH-Cond Meter / Get Thermometer headers are SMPL: Equipment Select.

Reuses: **NEW — iterative per-addition titration with running totals; no library block. Header uses SMPL: Equipment Select ×3**

▸ Get Scale / Balance

Scale / Balance ID

Scale / Balance Description

Scale / Balance Cal Due

▸ Get pH / Cond Meter

pH / Cond Meter ID

pH / Cond Meter Description

pH / Cond Meter Cal Due

▸ Get Thermometer

Thermometer ID

Thermometer Description

Thermometer Cal Due

Initial pH *

Initial Conductivity *

Initial Temperature (°C) *

Data Input

Addition #	Gross Wt Before (kg)	Gross Wt After (kg)	Net Added (kg)	Total Net (kg)		pH Sample Time	Adjusted pH	Sample Temp (°C)	Performed By *
1					▸ Record	⏱		⏱	

+ Add Row

▸ Record

Final pH Confirmed Time *

Final pH

Final Conductivity *

Final Temperature (°C) *

▸ Record

Incubation Start Time *

Witnessed By *

SMPL: Incubation Timing & Incubated Sample (block stack) REUSE

Original BR: §11 / 13

Time the low-pH incubation (start carries from the titration; target 75 min (NOR 60-240)), stamp the mid-incubation sample time, take a mid-incubation temperature check and an incubated pH/cond/temp sample, then make the §13 decision — in-range proceed to neutralization, out-of-range extend incubation and contact the supervisor. Assembled from the Timer block + a Record Text (sample time) + two Equipment Selects + the Solution Summary results block + a Dynamic Dropdown (drives conditional activation) + a Record Text (supervisor). No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Timer for Start-End Process (incubation start/end/duration) + SMPL: Record Text Value (mid-incubation sample time) + SMPL: Equipment Select ×2 (thermometer + pH/Cond meter) + SMPL: Solution Summary - Data Recording (incubated pH/cond/temp) + Dynamic Dropdown (§13 in-range → neutralize / out → extend incubation) + SMPL: Record Text Value (supervisor contacted on extension)**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Timer for Start-End Process — low-pH incubation (target 75 min, NOR 60–240)

Incubation Start Time (carries from titration)

▸ Record

Incubation End Time *

Incubation Duration

SMPL: Record Text Value — mid-incubation sample time

▸ Record

Mid-Incubation Sample Time *

SMPL: Equipment Select — Thermometer

▸ Get Thermometer

Thermometer ID

Calibration Due Date

SMPL: Equipment Select — pH / Conductivity Meter

▸ Get pH / Cond Meter

pH / Cond Meter ID

Calibration Due Date

SMPL: Solution Summary - Data Recording — incubated pH / conductivity / temperature

#	Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High / Low)
1	Incubated pH					

+ Add Row

§13 Decision — pH in range? * In-range *— activates / deactivates the section below*

In-range → proceed to neutralization. Out-of-range → extend incubation (re-time) and contact supervisor.

SMPL: Record Text Value — supervisor contacted (only if incubation extended)

Supervisor Contacted *

SMPL: Starting / Transfer Temperature Check (block stack) REUSE

Original BR: §7.14 / 9

Measure product temperature and record it. No Dynamic Dropdown is needed: the Record Numeric Value XStep carries Min/Max targets (18–25 °C) and its min/max validation function module enforces the range — an in-range value proceeds to acid addition, an out-of-range value triggers the configured error handling (warm the product if below range / contact supervisor if above). Assembled from Equipment Select + Record Numeric Value. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Equipment Select (thermometer) + SMPL: Record Numeric Value (product temperature, Min/Max 18–25 °C). No decision dropdown — the Record Numeric Value's Min/Max validation function module enforces the range and fires the configured error handling on out-of-range (warm if low / supervisor if high).**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Equipment Select — Thermometer

▸ Get Thermometer

Thermometer ID

Calibration Due Date

SMPL: Record Numeric Value — product temperature (IV_MIN/IV_MAX = 18–25 °C set on the step; Min/Max validation FM enforces the range + error messaging)

Product Temperature (°C) *

No decision dropdown: the Min/Max validation function module on the Record Numeric Value flags out-of-range and runs the configured error handling (below range → warm the product, Section 9; above range → contact supervisor); an in-range value proceeds to acid addition (Section 10).

SMPL: Store VI Product & Hold-Time Link (block stack) REUSE

Original BR: §15.8

Store the neutralized VI product and map values into Attachment 3. This is simply the Product Storage & Cross-Record step + the Hold Time Table, instantiated here — no separate XStep.

Reuses: **Stack of this record's steps: Common - Product Storage & Cross-Record + Common - Hold Time Table, instantiated for §15.8**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

Common - Product Storage & Cross-Record (instantiated for §15.8)

Storage Condition (RT / 2-8 °C) * 2-8 °C

▸ Record

Date *

Time *

Cross-Record — value written to Attachment 3 *

Common - Hold Time Table (instantiated for §15.8)

Hold #	Location	Storage Temp	Storage Start	Storage End	RT Hold (h)	2-8°C Hold (h)	Performed By *
1		2-8 °C					

+ Add Row

Concentration & Diafiltration — PN 8012475 (Large Skid UF/DF, 2000 L)

SMPL: Record CIPDS / Skid / Cassette Information (block stack) REUSE

Original BR: §7.2 / §11 / Att6

Record the UF/DF skid & membrane identity (Skid ID, CIPDS batch, cassette lot/part, support plate, hold-up volume, surface area), confirm the ≤30-day storage window, set up and label the Retentate Vessel — Tank ID, cleaning & sterilization batch/expiry, pressure-test date, vessel volume, vent filter — and capture the retentate tare before and after attachment. Assembled from Record Text Value ×n (collapse to a table) + Material Consumption + Select Vessel Type & Tare Weight (the vessel-setup block) + Equipment Select + Record Scale Weight + Record Numeric Value. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Record Text Value ×n (Skid ID / CIPDS Batch / Cassette Lot / Support Plate / sight-glass / valve labels) + SMPL: Material Consumption (cassette / CIPDS / Retentate Vessel vent filter) + SMPL: Select Vessel Type & Tare Weight (Retentate Vessel: Tank ID + cleaning Batch/Exp + sterilization Batch/Exp + pressure-test date + volume) + SMPL: Equipment Select (scale) + SMPL: Record Scale Weight (retentate tare before & after attachment, Scale ID/Cal Due) + SMPL: Record Numeric Value (hold-up volume / total surface area)**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Record Text Value ×n — skid & membrane identity (collapse to one table)

UF/DF Skid ID *

CIPDS Batch No. *

Cassette Install Date *

No. of Cassettes *

SMPL: Material Consumption — cassette / support plate / Retentate vent filter SAP Goods Issue · Z_PICONS mvt 261

#	Component	Part No.	Batch No.	Exp. Date	Serial No.	Performed By *
1	Membrane Cassette					

+ Add Row

SMPL: Record Numeric Value — hold-up volume & total surface area

Hold-Up Volume (L) *

Total Membrane Surface Area (m²) *

Cassette storage ≤ 30 days? * Yes — *activates / deactivates the section below*

Confirm the ≤30-day cassette storage window before use.

SMPL: Select Vessel Type & Tare Weight — Retentate Vessel set-up (§11)

Tank ID *

Cleaning Batch / Exp. *

Sterilization Batch / Exp. *

Pressure Test Date *

Vessel Volume (L) *

SMPL: Equipment Select — Scale / Balance

▸ Get Scale / Balance

Scale / Balance ID

Calibration Due Date

SMPL: Record Scale Weight — Retentate tare (before & after attachment)

Tare Weight before attachment (kg) *

Tare Weight after attachment (kg) *

SMPL: Run Skid Recipe (block stack) REUSE

Original BR: §8/10/13/15/19

Download and run the named skid recipe per phase (Prep-WFI Flush, Equilibration, Conc 1, Diafiltration, Conc 2, Recovery, Clean-WFI Flush, Sanitization): record the Batch ID and recipe setpoints, stamp recipe start and any ≥5-min recirculation (Timer), record retentate weights (Record Scale Weight) and permeate/retentate totalizers, capture the retentate/permeate conductivity & temperature sample results (Solution Summary), log in-process parameters (Operation Monitoring Worksheet) and take the conductivity in-range / re-flush decision (Dynamic Dropdown). WFI-flush phases (§8/§19) first set up WFI — record Bag / Valve / Tank ID and Charge Date/Time and compute the 24-h expiry. Buffer / WFI / NaOH consumed post via Goods Issue (Z_PICONS 261). No bespoke XStep — every field maps to an existing block.

Reuses: **DE1 100 stack: SMPL: Record Text Value (Batch ID + recipe setpoints TMP/crossflow/flush-vol) + SMPL: Record Numeric Value (membrane-area setpoint, permeate/retentate totalizers) + SMPL: Timer for Start-End Process (recipe start + ≥5-min recirculation start/end/duration) + SMPL: Record Scale Weight (retentate weights + Scale ID/Cal Due) + SMPL: Solution Summary - Data Recording (retentate/permeate conductivity & temperature results) + CDF - Operation Monitoring Worksheet (in-process TMP/pressure/flow/UV log) + Dynamic Dropdown (WFI re-flush / conductivity in-range decision) + SMPL: Component Goods Issue (8012555 buffer / WFI / NaOH per phase); §8/§19 add WFI set-up: SMPL: Record Text Value (Bag / WFI Valve-Tank ID + Charge Date/Time) + SMPL: Calc Three Columns (Expiration = Charge + 24 h)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

WFI-flush phase? (§8 / §19) * No — *activates / deactivates the section below*

WFI-flush phases (§8/§19) first set up WFI below; other phases skip to the recipe.

SMPL: Record Text Value — WFI set-up (WFI-flush phases only)

Bag / WFI Valve-Tank ID *

▸ Record

WFI Charge Date & Time *

SMPL: Calc Three Columns — WFI Expiration = Charge + 24 h

Value 1		Value 2		Result
WFI Charge Date/Time	+	24 h	=	WFI Expiration Date/Time

+ Add Row

SMPL: Record Text Value — recipe batch & setpoints

Recipe / Batch ID *

Setpoints (TMP / Crossflow / Flush Vol) *

SMPL: Timer for Start-End Process — recipe start & ≥5-min recirculation

▸ Record

Recipe Start Time *

▸ Record

Recirculation Start *

▸ Record

Recirculation End *

Recirculation Duration (≥ 5 min)

SMPL: Record Numeric Value — membrane-area setpoint & totalizers

Membrane Area Setpoint (m^2) *

Permeate Totalizer (L) *

Retentate Totalizer (L) *

SMPL: Record Scale Weight — retentate weight

▸ Get Scale / Balance

Scale ID

Calibration Due Date

Retentate Weight (kg) *

SMPL: Solution Summary - Data Recording — retentate / permeate conductivity & temperature

#	Stream / Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High/Low)
1	Permeate Conductivity					

+ Add Row

CDF - Operation Monitoring Worksheet — in-process TMP / pressures / flow / UV log (its own Att-2 card)

Conductivity in range? / WFI re-flush? * In-range — *activates / deactivates the section below*

In-range → proceed. Out-of-range → re-flush and re-sample.

SMPL: Component Goods Issue — buffer / WFI / NaOH consumed this phase SAP Goods Issue · Z_PICONS mvt 261

#	Component	Batch No.	Amount Used (L)	Performed By *
1	Buffer 8012555			

+ Add Row

SMPL: Minimum Membrane Surface Area (block stack) REUSE

Original BR: §7.7

Total Grams Product ÷ Max Loading (g/m²) = Minimum Membrane Area; confirm installed area ≥ minimum. Assembled from a 2-input Calc Three Columns + Record Numeric Value + a Dynamic Dropdown. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Calc Three Columns (Total Grams ÷ Max Loading = Min Area) + SMPL: Record Numeric Value (installed area) + Dynamic Dropdown (installed ≥ minimum?)**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Calc Three Columns — Minimum Area = Total Grams ÷ Max Loading

Value 1		Value 2		Result
Total Grams Product	÷	Max Loading (g/m²)	=	Minimum Membrane Area (m²)

+ Add Row

SMPL: Record Numeric Value — installed membrane area

Installed Membrane Area (m²) *

Installed area ≥ minimum? * Yes — *activates / deactivates the section below*

Confirm the installed membrane area ≥ calculated minimum before proceeding.

SMPL: Process Input Calculations (calc-block stack) REUSE

Original BR: §12

Compute the process setpoints from the load (capacity 80%/40%, Conc 1 & 2 target volumes/weights, DF permeate range). A stack of ~9 reusable calc blocks (2-input Calc Three Columns, 3-input Three Variable Calc, and Calc Range Values for the ranges). No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Calc Three Columns / Three Variable Calc / Calc Range Values (×~9 setpoint calcs — 80%/40% capacity, Conc 1/2 volumes & weights, DF permeate range)**

▼ *Representative* **PI Sheet view** *— this is a* **block stack** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Calc stack (~9) — 80%/40% capacity, Conc 1 & 2 volumes/weights, DF permeate range

#	Calculation	Value 1		Value 2		Result (Min / Target / Max)
1	Vessel charge @ 40% capacity	Vessel Volume	×	0.40	=	Charge Volume / Weight

+ Add Row

SMPL: UF/DF Operation Monitoring Worksheet (Att 2) NEW

Original BR: Att2 (§13-15)

Instructions

Log operating parameters during Concentration 1 / Diafiltration / Concentration 2 using the same conductivity meter as set-up (Step 8.2), including temperature compensation. Per reading record the three transducer pressures (Feed PT-101 / Retentate PT-301 / Permeate PT-201) and computed ΔP, the retentate back-pressure valve opening (BPRV-301) and pump speed (P-101), crossflow flux, TMP (TMP-100), permeate flow (FT-201), permeate flux, total permeate volume (FQI-201), permeate UV (AIT-201), offline permeate conductivity & pH, retentate agitator setting and product weight; the max feed pressure alarm (≤ 60 psig) must not be exceeded. Built new — no library block covers this wide interval log.

Reuses: **NEW — wide interval log (TMP / 3 pressures / permeate flow / conductivity / vessel weight); no library block (Timer-table too narrow, generic parameterized table is a demo). Get Conductivity Meter = SMPL: Equipment Select**

▸ Get Conductivity Meter

Conductivity Meter ID

Conductivity Meter Description

Conductivity Meter Cal Due

Data Input

#		Time	PFeed (psig) PT-101	PRetentate (psig) PT-301	PPermeate (psig) PT-201	ΔP (psid)	Retentate Open (%) BPRV-301	Pump Speed (RPM) P-101	Crossflow (L/min/ft²)	TMP (psid) TMP-100	Permeate Flow (L/min) FT-201	Permeate Flux (LMH)	Total Permeate Vol (L) FQI-201	Permeate UV (OD) AIT-201	Permeate Cond. (mS/cm) *	Permeate pH *	Agitator Setting	Product Wt (kg)	Performed By *
1	▸ Record	⏱																	

+ Add Row

Witnessed By *

SMPL: Conditional Continued Diafiltration (block stack) REUSE

Original BR: §14

Run the diafiltration recipe and log parameters, record the net weight and compute product volume, then sample the permeate/retentate (pH/cond/temp); if in range proceed, else contact the supervisor, run additional diavolumes and re-sample. Additional 8012555 buffer posts via Goods Issue (Z_PICONS 261). Assembled from the Run Skid Recipe stack + Operation Monitoring + Record Scale Weight + Calc + Equipment Select + Solution Summary + a Dynamic Dropdown + Record Numeric + Record Text. (The decision was published to the PI Sheet Sandbox as a flat form — that stays a representation.) No bespoke XStep.

Reuses: **DE1 100 stack: CDF - Run Skid Recipe (DF1244_Diafiltration batch ID / start / setpoints) + CDF - Operation Monitoring Worksheet (in-process log) + SMPL: Record Scale Weight (net weight after DF + Scale ID/Cal Due) + SMPL: Calc Three Columns (product volume = net wt ÷ density) + SMPL: Equipment Select (pH/Cond meter) + SMPL: Solution Summary - Data Recording (permeate/retentate pH/cond/temp) + Dynamic Dropdown (Continue?) + SMPL: Record Numeric Value (additional diavolumes / permeate totalizer) + SMPL: Record Text Value (supervisor contacted on out-of-range)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

CDF - Run Skid Recipe — DF1244_Diafiltration (batch ID / start / setpoints) — see Run Skid Recipe card

CDF - Operation Monitoring Worksheet — in-process log (its own Att-2 card)

SMPL: Record Scale Weight — net weight after Diafiltration

▸ Get Scale / Balance

Scale ID

Calibration Due Date

Net Weight after DF (kg) *

SMPL: Calc Three Columns — Product Volume = Net Weight ÷ Density

Value 1		Value 2		Result
Net Weight after DF	÷	Density	=	Product Volume (L)

+ Add Row

SMPL: Equipment Select — pH / Conductivity Meter

▸ Get pH / Cond Meter

pH / Cond Meter ID

Calibration Due Date

SMPL: Solution Summary - Data Recording — permeate / retentate pH / cond / temp

#	Stream / Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High/Low)
1	Permeate pH					

+ Add Row

Continue Diafiltration? (pH in range?) * In-range *— activates / deactivates the section below*

In-range → proceed to Concentration 2. Out-of-range → contact supervisor, run additional diavolumes and re-sample.

SMPL: Record Numeric Value — additional diavolumes / permeate totalizer

Additional Diavolumes *

Permeate Totalizer (L) *

SMPL: Record Text Value — supervisor contacted (out-of-range)

Supervisor Contacted *

SMPL: Recovery Calculations & Execution (block stack) REUSE

Original BR: §16

Compute the recovery target, choose method (manual/recipe), recover product to the Dilution Vessel, then transfer & mix the recovered product: record the transfer tubing (with hose-cleaning info) and the 'Mixer Drive attached at tare?' flag, stamp transfer end, mix ≥20 min, pull the 8012475-20 sample, weigh the vessel (gross/tare/net) and record concentration + DLIMS, then decide whether dilution is required. Label the PFI mixing bag (SOP-0107056). Recovery buffer posts via Goods Issue (Z_PICONS 261). Assembled from calc blocks + Equipment Select + Material Consumption + Dynamic Dropdown(s) + Run Skid Recipe + Transfer Tubing + Timer + Mixing Time + Sampling + Record Scale Weight + Record Numeric/Text. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Calc Three Columns / Three Variable Calc / Calc Range Values (recovery volume/weight range, buffer amount, net/volume after transfer) + SMPL: Equipment Select (mixer + scale) + SMPL: Material Consumption (recovery buffer + Dilution Vessel bag) + Dynamic Dropdown (method manual/recipe + 'Mixer Drive attached at tare?' Yes/No + dilution-required?) + CDF - Run Skid Recipe (recipe-control path) + Common - Transfer Tubing and Hosing Info (transfer line + Hose Cleaning) + SMPL: Timer for Start-End Process (transfer end) + SMPL: Mixing Time (Dilution Vessel mix ≥20 min) + SMPL: Sampling Record + SMPL: Sample Submission Chart (8012475-20) + SMPL: Record Scale Weight (gross/tare/net + Scale ID/Cal Due) + SMPL: Record Numeric Value + SMPL: Record Text Value (supervisor contacted)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Calc stack — recovery volume/weight range, buffer amount, net/volume after transfer

#	Calculation	Value 1		Value 2		Result (range)
1	Recovery Product Volume Range		×		=	

+ Add Row

Recovery Method? * By recipe control — *activates / deactivates the section below*

Manual → skip the recipe fields. By recipe control → run the recovery recipe below.

CDF - Run Skid Recipe — recovery recipe (recipe-control path; see Run Skid Recipe card)

SMPL: Equipment Select — mixer + scale

▸ Get Mixer

Mixer ID

▸ Get Scale / Balance

Scale ID

Calibration Due Date

SMPL: Material Consumption — recovery buffer + Dilution Vessel bag SAP Goods Issue · Z_PICONS mvt 261

#	Component	Part No.	Batch No.	Exp. Date	Performed By *
1	Recovery Buffer 8012555				

+ Add Row

Mixer Drive attached at tare? * Yes / No

Common - Transfer Tubing and Hosing Info — transfer line + hose cleaning (see Transfer Tubing card)

SMPL: Timer + SMPL: Mixing Time — transfer end & Dilution Vessel mix (≥ 20 min)

▸ Record

Transfer End Time *

▸ Record

Mixing Start *

▸ Record

Mixing End *

Mixing Duration (≥ 20 min)

SMPL: Sampling Record + Sample Submission Chart — 8012475-20

Sample Designation 8012475-20

▸ Record

Sampling Date & Time *

DLIMS Project / Sample No. *

SMPL: Record Scale Weight — Dilution Vessel gross / tare / net

Gross Weight (kg) *

Tare Weight (kg)

Net Weight (kg)

SMPL: Record Numeric Value — recovered product concentration + DLIMS

Concentration (g/L) *

DLIMS Project ID *

DLIMS Sample ID *

Dilution Required? * No — *activates / deactivates the section below*

Based on the 8012475-20 concentration → dilution required (Section 17) or not.

SMPL: Record Text Value — supervisor contacted (if applicable)

Supervisor Contacted *

SMPL: Dilution Decision & Calculations (block stack) REUSE

Original BR: §17

Decide whether dilution is required (target ~180 g/L PFI); if so compute the dilution buffer amount, obtain the Intermediate Vessel & transfer tubing, add buffer to target weight (Goods Issue Z_PICONS 261), mix (≥20 min then ≥15 min), pull the -30/-40 samples and record pH/conductivity/temperature, weigh and compute PFI net weight/volume; label the Intermediate Vessel (SOP-0107056). Assembled from Dynamic Dropdown(s) + calc blocks + three Equipment Selects + Material Consumption + Transfer Tubing + Record Scale Weight + Mixing Time ×2 + Sampling + Solution Summary + Record Numeric/Text. No bespoke XStep.

Reuses: **DE1 100 stack: Dynamic Dropdown (dilution required? + dilution method + 'was there dilution?') + SMPL: Calc Three Columns / Calc Range Values (grams, PFI volume/weight range, buffer weight & volume, net/volume after sampling) + SMPL: Equipment Select ×3 (pump / stir plate / scale) + SMPL: Material Consumption (dilution buffer + Intermediate Vessel) + Common - Transfer Tubing and Hosing Info (buffer transfer line) + SMPL: Record Scale Weight (net buffer added, gross after attachments + Scale ID/Cal Due) + SMPL: Mixing Time ×2 (mix ≥20 min then ≥15 min) + SMPL: Sampling Record + SMPL: Sample Submission Chart (8012475-30 / -40) + SMPL: Solution Summary - Data Recording (pH/conductivity/temperature) + SMPL: Record Numeric Value (PFI concentration result) + SMPL: Record Text Value (supervisor contacted)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

Dilution Required? * No — *activates / deactivates the section below*

Based on recovered-product concentration. No → N/A the dilution steps. Yes → perform dilution below.

SMPL: Calc stack — grams, PFI volume/weight range, buffer weight & volume

#	Calculation	Value 1		Value 2		Result (range)
1	Weight of Dilution Buffer to Add		×		=	

+ Add Row

Dilution Method? * Connect buffer to vessel — *activates / deactivates the section below*

Pipette / cylinder / carboy → use an Intermediate Vessel. Direct → connect buffer to the Dilution Vessel.

SMPL: Equipment Select ×3 — pump / stir plate / scale

▸ Get Pump

Pump ID

▸ Get Stir Plate

Stir Plate ID

▸ Get Scale / Balance

Scale ID

Calibration Due Date

SMPL: Material Consumption — dilution buffer + Intermediate Vessel SAP Goods Issue · Z_PICONS mvt 261

#	Component	Part No.	Batch No.	Exp. Date	Performed By *
1	Dilution Buffer 8012555				

+ Add Row

Common - Transfer Tubing and Hosing Info — buffer transfer line (see Transfer Tubing card)

SMPL: Record Scale Weight — buffer added / gross after attachments

Net Buffer Added (kg) *

Gross after attachments (kg) *

Scale ID

Calibration Due Date

SMPL: Mixing Time ×2 — mix ≥ 20 min then ≥ 15 min

Mix 1 Duration (≥ 20 min)

Mix 2 Duration (≥ 15 min)

SMPL: Sampling Record + Sample Submission Chart — 8012475-30 / -40

Sample Designation: 8012475-30 / -40

▸ Record

Sampling Date & Time *

DLIMS Project / Sample No. *

SMPL: Solution Summary - Data Recording — pH / conductivity / temperature

#	Parameter	Equipment	Min Target	Max Target	Actual Value	Cond (High/Low)
1	pH					

+ Add Row

SMPL: Record Numeric Value — PFI concentration result

PFI Concentration (g/L) *

SMPL: Record Text Value — supervisor contacted (if out of range)

Supervisor Contacted *

SMPL: PFI Storage Vessel, SHC Filter & Store (block stack) REUSE

Original BR: §18

Obtain & label the PFI storage vessel (bag or autoclaved glass bottle) and SHC filter, connect the transfer tubing, transfer the diluted product through the filter, stamp transfer end, weigh (tare + gross) and compute net weight/volume, then label and store and cross-record hold times into Attachment 5. Assembled from Storage Vessel and Filter Inform + Additional Assembly (bottle) + Material Consumption + Transfer Tubing + Equipment Select + Record Scale Weight + Timer + Calc + Record Numeric + Solution Final Storage + Hold Time Table. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Storage Vessel and Filter Inform (storage vessel + air/vent filter — field-verified) + SMPL: Additional Assembly (glass-bottle autoclave cycle/exp, if bottle) + SMPL: Material Consumption (SHC filter) + Common - Transfer Tubing and Hosing Info (transfer line + Hose Cleaning) + SMPL: Equipment Select (scale) + SMPL: Record Scale Weight (tare + gross after transfer) + SMPL: Timer for Start-End Process (transfer end) + SMPL: Calc Three Columns (net = gross − tare; volume = net ÷ density) + SMPL: Record Numeric Value (net weight) + SMPL: Solution Final Storage + Common - Hold Time Table (Att5 cross-record)**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Storage Vessel and Filter Inform — PFI storage vessel + air/vent filter

Storage Vessel (Bag / Bottle) Part / Batch / Exp. *

Air / Vent Filter Part / Batch / Exp. *

Glass bottle used? * No — *activates / deactivates the section below*

If a glass bottle, record its autoclave cycle / expiry below (SMPL: Additional Assembly).

SMPL: Additional Assembly — glass-bottle autoclave (if bottle)

Glass Bottle Autoclave Cycle No. *

Autoclave Exp. Date *

SMPL: Material Consumption — SHC product filter SAP Goods Issue · Z_PICONS mvt 261

Filter	Part No.	Batch No.	Exp. Date	Autoclave Cycle No.	Autoclave Exp.	Performed By *
PFI Product Filter (SHC)						

Common - Transfer Tubing and Hosing Info — transfer line + hose cleaning (see Transfer Tubing card)

SMPL: Equipment Select + Record Scale Weight — tare & gross after transfer

▸ Get Scale / Balance

Scale ID

Calibration Due Date

Tare Weight (kg) *

Gross after transfer (kg) *

SMPL: Timer for Start-End Process — transfer end

▸ Record

Transfer End Time *

SMPL: Calc Three Columns — net = gross − tare; volume = net ÷ density

#	Value 1		Value 2		Result
1	Gross Weight	-	Tare Weight	=	Net Weight (kg)

+ Add Row

SMPL: Solution Final Storage — store PFI

Storage Location *

▸ Record

Date *

Time *

Common - Hold Time Table — cross-record to Attachment 5 (see Hold Time card)

SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack) REUSE

Original BR: §19 / §20

Set up WFI (Bag / Valve / Tank ID + Charge Date/Time, 24-h expiry computed), confirm the conductivity meter, enter the membrane surface-area setpoint, run the post-use DF1244_Flush (Clean-WFI), then decide cassette re-use vs discard and run the NaOH sanitization / storage recipe (NaOH via Goods Issue Z_PICONS 261), recording logbook/CIPDS batch, action dates and the buffer batch & amount used (totalizer sum). Assembled from Record Text + Calc + Equipment Select + Record Numeric + Run Skid Recipe + Dynamic Dropdown + Material Consumption + Timer + Text Instructions. No bespoke XStep.

Reuses: **DE1 100 stack: SMPL: Record Text Value (WFI Bag / Valve-Tank ID + Charge Date/Time) + SMPL: Calc Three Columns (WFI Expiration = Charge + 24 h) + SMPL: Equipment Select (conductivity meter) + SMPL: Record Numeric Value (membrane-area setpoint, permeate/retentate flush setpoints, totalizer amount used) + CDF - Run Skid Recipe (DF1244_Flush batch ID / start) + Dynamic Dropdown (cassette re-use vs discard) + SMPL: Material Consumption (NaOH sanitization / storage buffer) + SMPL: Record Text Value (logbook / CIPDS batch + sanitization/NWP/storage dates + buffer batch) + SMPL: Timer for Start-End Process (start) + SXS: Text Instructions with Sign-off**

▼ *Representative* ***PI Sheet view*** *— this is a* ***block stack*** *(no new object), assembled from the reused DE1 100 blocks named above; each labelled section / table below is one of those blocks as it renders in the EBR:*

SMPL: Record Text Value — WFI set-up

Bag / WFI Valve-Tank ID *

▸ Record

WFI Charge Date & Time *

SMPL: Calc Three Columns — WFI Expiration = Charge + 24 h

Value 1		Value 2		Result
WFI Charge Date/Time	+	24 h	=	WFI Expiration Date/Time

+ Add Row

SMPL: Equipment Select — conductivity meter

▸ Get Conductivity Meter

Meter ID

Calibration Due Date

SMPL: Record Numeric Value — membrane-area & flush setpoints

Membrane Area Setpoint (m^2) *

Permeate Flush Setpoint (L) *

Retentate Flush Setpoint (L) *

CDF - Run Skid Recipe — DF1244_Flush (Clean-WFI) batch ID / start (see Run Skid Recipe card)

Cassettes re-used? * Discard — *activates / deactivates the section below*

Discard → NWP not required; clean & store the skid. Re-use → clean, WFI flush, NWP and store.

SMPL: Material Consumption — NaOH sanitization / storage buffer SAP Goods Issue · Z_PICONS mvt 261

#	Buffer	Batch No.	Amount Used (L)	Performed By *
1	0.5 N NaOH (Sanitization)			

+ Add Row

SMPL: Record Text Value — logbook / CIPDS batch + action dates

Logbook No. *

CIPDS Batch No. *

Sanitization / NWP / Storage Dates *

SMPL: Timer for Start-End Process — start

▸ Record

Start Time *

SXS: Text Instructions with Sign-off — BPR review / post-processing (§20)